

M.Tech. (Mining Engineering) Semester- I, Midterm Examination - 2024

Subject: MN572, Reliability Engineering

Time: 2 Hours

Marks: 30

0-256

Q.1 (a) 'Component - X' has a linear hazard rate function given by: (2)

$$\lambda(t) = 0.4t \quad \text{when failure times (t) in years}$$

- Determine the probability of 'component - X' failing within the first month of its operation.
- What is the design life of the 'component - X' for 0.95 desired reliability?

(b) The PDF for the time to failure in days for a system is given by (5)

$$f(t) = 0.01 \quad 0 \leq t \leq 100 \text{ days}$$

- Calculate the probability that the component will not fail within 60 days of operation
- What is the failure rate after 60 days of operation?
- Find the MTTF.
- Find the median time to failure.
- Compute the standard deviation of failure times.

(c) A new fuel injection system is experiencing high failure rates. The reliability function has been found to be (3)

$$R(t) = (t + 1)^{-\frac{3}{2}} \quad \text{for } t \geq 0 \text{ and } t \text{ is measured in years}$$

Will a burn-in period of 6 months significantly improve the fuel injection system's reliability over its designed life of 2 years?

Q. 2 (a) Define and explain the following terms: (4)

- (i) Reliability (ii) Availability (iii) Maintainability
(iv) Primary and secondary effects of failure

(b) What are the objectives of reliability engineering? (2)

(c) Discuss the modes, causes and mechanisms of failures. (2)

(d) Discuss various reliability and availability improvement methods of a system during its design and operational phases. (2)

Q. 3 (a) The pumping system of a mine consists of two independent and identical submersible pumps. Each pump has the same constant failure rate λ and operates in a **redundant configuration**. Derive the expressions for $\lambda(t)$ and MTTF for the pumping system. (2)

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(b) A component experiences 0.001 failures per hour. It has a guaranteed life of 200 hours. (5)

- i. Find the reliability at 300 hours of operation.
- ii. Calculate MTTF and σ of failure times.
- iii. What is the median life of the component?
- iv. Calculate the designed life if a 0.95 reliability is desired.

(c) A compressor experiences wear-out with a linear hazard rate function as given below: (3)

$$\lambda(t) = 2 \times 10^{-6}t$$

What are the values of the shape parameter (β) and scale parameter (θ)? Calculate the designed life for a reliability of 0.99, MTTF and σ of failure times. Take $\Gamma(1.5) = 0.88623$.

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